AI LISTENER

Brydge

A Conversational AI Psychologist

Academic · Peer · Family

Free | Open Source | Apache 2.0 | Mistral 7B | Flask | 220+ Frameworks

"Because every voice deserves to be heard."

Healing starts by being heard · Built with Python · Flask · HuggingFace · Mistral 7B MH Fine-tuned

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Introduction

What is Brydge?

Brydge is a free, web-based AI-powered mental health support platform for young people. It uses a conversational AI psychologist — the Listener — to provide empathetic, evidence-based support around academic, peer, and family challenges, 24 hours a day at zero cost.

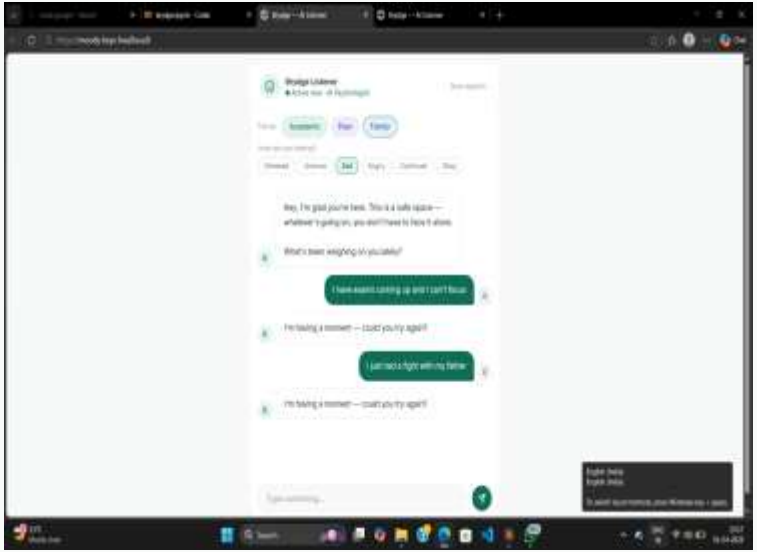
£0
Cost to users

24/7
AI available

220+
Frameworks loaded

3
Life spheres

Live Demo Screenshot



Three Spheres & Associated Frameworks

Academic

Issues Covered:

Exam anxiety · Procrastination · Focus problems · Perfectionism · Burnout · Imposter syndrome · Teacher conflict · Academic pressure

Frameworks Used:

- CBT (Beck)
- SFBT (de Shazer)
- Behavioural Activation
- Growth Mindset (Dweck)
- WOOP
- ACT (Hayes)
- Anxiety Reappraisal

Peer Issues

Issues Covered:

Bullying · Social exclusion · Loneliness · Friendship breakdown · Social anxiety · Peer pressure · Online harassment · Romantic rejection

Frameworks Used:

- DBT — DEAR MAN (Linehan)
- NVC (Rosenberg)
- Attachment Theory (Bowlby)
- Narrative Therapy (White)
- Assertiveness Training
- CBT for Social Anxiety

Family Issues

Issues Covered:

Parent conflict · Divorce · High expectations · Feeling unheard · Cultural pressure · Financial stress · Role reversal · Communication breakdown

Frameworks Used:

- Family Systems (Satir)
- Structural Therapy (Minuchin)
- Bowen Theory
- Emotion Coaching (Gottman)
- NVC
- IPT (Klerman)
- Contextual Therapy

220+ Therapeutic Frameworks — 8 Core Psychologists

FRAMEWORK CATEGORIES

→ 22 frameworks	Cognitive (CBT, REBT, MCT, CPT, Schema)
→ 18 frameworks	Acceptance & Mindfulness (ACT, DBT, MBCT, MBSR, CFT)
→ 16 frameworks	Humanistic & Existential (PCT, Gestalt, Logotherapy, Existential)
→ 12 frameworks	Solution-Focused (SFBT, Possibility, Solution-Oriented)
→ 14 frameworks	Interpersonal & Relational (IPT, Attachment, TA, CCRT)
→ 14 frameworks	Family Systems (Bowen, Structural, Strategic, Satir, Milan)
→ 12 frameworks	Motivational (MI, Stages of Change, SDT, Self-Efficacy)
→ 16 frameworks	Trauma-Informed (EMDR, SE, NARM, Polyvagal, IFS)
→ 16 frameworks	Psychodynamic (Freud, Jung, Adler, Klein, Winnicott, Kohut)
→ 10 frameworks	Communication (OARS, NVC, Active Listening, Egan's Model)
→ 20 skills	DBT Skills (TIPP, DEAR MAN, GIVE, FAST, ACCEPTS, IMPROVE)
→ 18 techniques	Positive Psychology (PERMA, VIA, Gratitude, Growth Mindset)
→ 14 techniques	Grounding & Stabilisation (5-4-3-2-1, Box Breathing, Body Scan)

Total: 220+ frameworks, techniques & skills embedded in system prompt

8 CORE PSYCHOLOGISTS

Aaron Beck CBT	Negative thoughts → cognitive restructuring
Carl Rogers Person-Centred	Unconditional positive regard, empathy
Marsha Linehan DBT	Emotion regulation, DEAR MAN, radical acceptance
Steven Hayes ACT	Defusion, values clarification, committed action
William Miller MI — OARS	Open questions, reflections, no pushing
Virginia Satir Family Systems	Reframe conflict as systemic, not personal
Steve de Shazer SFBT	Miracle question, exception finding, scaling
Albert Ellis REBT	Challenge irrational 'MUST' beliefs

Problem Statement — What We're Solving

No Support After Hours

Mental health crises happen at 2am. Clinics are closed. No free 24/7 option exists.

Robotic Apps

Existing apps feel scripted & clinical. Users disengage after 1–2 sessions.

Wasted Peer Resource

Psychology students have empathy & training but no supervised real-world practice space.

Fear & Stigma

Most young people won't tell a teacher, parent, or peer they're struggling.

Cost Barrier

Professional therapy costs ₹1,500–₹4,000/session. Completely inaccessible to students.

No Evidence Base

Most free tools ignore established therapy frameworks like CBT, ACT, DBT, and MI.

Our Solution — Brydge AI Listener

24/7 AI Listener

Always-on conversational psychologist responds at any hour, any day.

Evidence-Based

220+ therapy frameworks from 8 psychologists embedded directly in the AI.

Free Forever

Apache 2.0 licence. No subscription. No credit card. No hidden fees.

Human Warmth

Warm, empathetic responses — not scripted. Feels like talking to a wise friend.

Peer Counsellors

Psychology students provide supervised real sessions. Adds human connection.

Safety First

Crisis detection layer intercepts danger signals before the AI ever responds.

Our Model — Mistral 7B Mental Health Fine-tuned

Model Card (from our comparison research)

★ Sweet spot pick

Mistral 7B MH

GRMenon/mental-health-mistral-7b

Quality	8.6/10
RAM needed	~6GB 4-bit
Empathy	High

+ Already fine-tuned on real psychologist-patient conversations
+ No access gate on HuggingFace
+ Mistral 7B base rated "closest to GPT-3.5" by community
+ Apache 2.0 licence — fully free
+ 4-bit quantised = runs on 8GB RAM

Technical Specifications

Model	GRMenon/mental-health-mistral-7b-instructv0.2-finetuned-V2
Base model	Mistral-7B-Instruct-v0.2 (7 billion parameters)
Fine-tuned on	Real psychologist–patient conversation dataset
Licence	Apache 2.0 — free commercial & research use
Access gate	None — downloads instantly from HuggingFace
Quantisation	4-bit (BitsAndBytes) — ~6GB RAM on laptop

Generation Parameters

`max_new_tokens` = 180	Enforces 4-sentence therapeutic rule. Avoids overwhelming the user.
`temperature` = 0.72	Natural enough to feel human. Stable enough to stay safe. Sweet spot.
`top_p` = 0.92	Nucleus sampling — diverse but controlled vocabulary.
`repetition_penalty` = 1.15	Prevents the model repeating the same phrases in long sessions.
`do_sample` = True	Allows varied, non-deterministic responses. More human-feeling.
`max_length (input)` = 3072 tokens	Full session history fits in one context window. No memory loss.

Why Mistral 7B MH over MentalLLaMA?

MentalLLaMA = 14GB + gated access + needs GPU · Mistral 7B MH = 6GB + no gate + runs on laptop

Libraries Used & Why These Specifically

HuggingFace Transformers

Load & run Mistral 7B in 3 lines of Python. Industry standard with 200k+ models. Active community.

Why not raw PyTorch? Too complex for a 2nd-year BTECH student. Transformers abstracts all of it.

Flask + Flask-SocketIO

Minimal Python web server. Full chatbot backend in under 200 lines. WebSocket chat in 10 lines.

Why not Django? Overkill — Django needs 10× the setup for our MVP. Flask is beginner-perfect.

BitsAndBytes (4-bit quant)

Compresses the 14GB Mistral model to ~6GB RAM so it runs on a standard laptop with no GPU.

Why this? Only library that supports 4-bit QLoRA quantisation out-of-box with one config line.

SQLite

Zero-setup database. Built into Python — no install, no server, no config. One file stores everything.

Why not PostgreSQL? SQLite handles our load perfectly at MVP stage. Switch later with one config change.

Tailwind CSS (CDN)

Professional, modern UI with a single <script> tag. No build tools, no Node.js setup needed.

Why not Bootstrap? Tailwind's utility classes give more control and a more modern look out of the box.

localtunnel

Creates a public HTTPS URL for the Flask app running in Google Colab in one line of code.

Ideal for development and testing. Free tier is enough for testing.

Data Structures & Why

List of Dictionaries — Conversation History

```
[
  {"role":"system",
"content":"...prompt..."},
  {"role":"user",
"content":"I can't focus"},

{"role":"assistant","content":"
That sounds hard..."}
]
```

Preserves message ORDER — critical for context.
O(1) append per message.
Required format for all HuggingFace chat models.

JSON Dictionary — Situation Database

```
{
  "id":       "AC001",
  "sphere":   "academic",
  "triggers":["can't
focus","blank mind"],
  "framework":"CBT",
  "psychologist":"Aaron Beck"
}
```

Human-readable — editable by psychology students.
No extra library needed (Python's json is built-in).
Fast O(1) key access per situation.

Dictionary — Judge LLM Scores

```
{
  "empathy":      8,
  "safety":       10,
  "framework":    7,
  "helpfulness": 8,
  "flag":         false,
  "overall":      8.4
}
```

O(1) key lookup for threshold decisions.
if scores['safety'] < 6: escalate immediately.
Simple to serialize and save as RLHF training log.

Dict + Nested List — User Session Store

```
sessions = {
  "user_abc123": {
    "sphere":   "academic",
    "severity": 3,
    "history":  [...messages],
    "started":  "2025-01-01"
  }
}
```

O(1) lookup by user_id — instant regardless of
how many users are active simultaneously.
Nested list (history) appends in O(1) per turn.

Data Characteristics

Data Type	Source	Format	Characteristics
User messages	Live input	Text string	Short 1–3 sentences · emotional · informal language · varies per user
Conversation history	Session RAM	List[Dict]	Grows each turn · max 20 turns before reset · ordered chronologically
Situation entries	Team curated	JSON file	1,000 entries · 3 spheres · static dataset · trigger-mapped to frameworks
AI responses	Mistral 7B model	Text string	Max 180 tokens · temperature 0.72 · top_p 0.92 · 4 sentences max
Judge LLM scores	Llama 3.2 3B	JSON dict	5 float scores 0–10 · 1 boolean flag · logged for RLHF training
Crisis signals	Keyword classifier	Boolean	Deterministic · runs BEFORE model · never misses · instant response
Session metadata	Flask server	Python dict	Per-user · auto-cleared after 45 min · sphere + severity + history
Training data	ESConv + CounselChat	JSONL	25k+ therapist–patient pairs · mental health domain · open licence

System Architecture

App PY	Open in Visual Studio Code
Listener PY	Open in Visual Studio Code
Chat Code · HTML	Open in Microsoft Edge
Colab setup PY	Open in Visual Studio Code
Requirements TXT	Open in Notepad
Readme Document · MD	Open in Visual Studio Code

Project Structure & How to Build and Run Brydge

File Structure

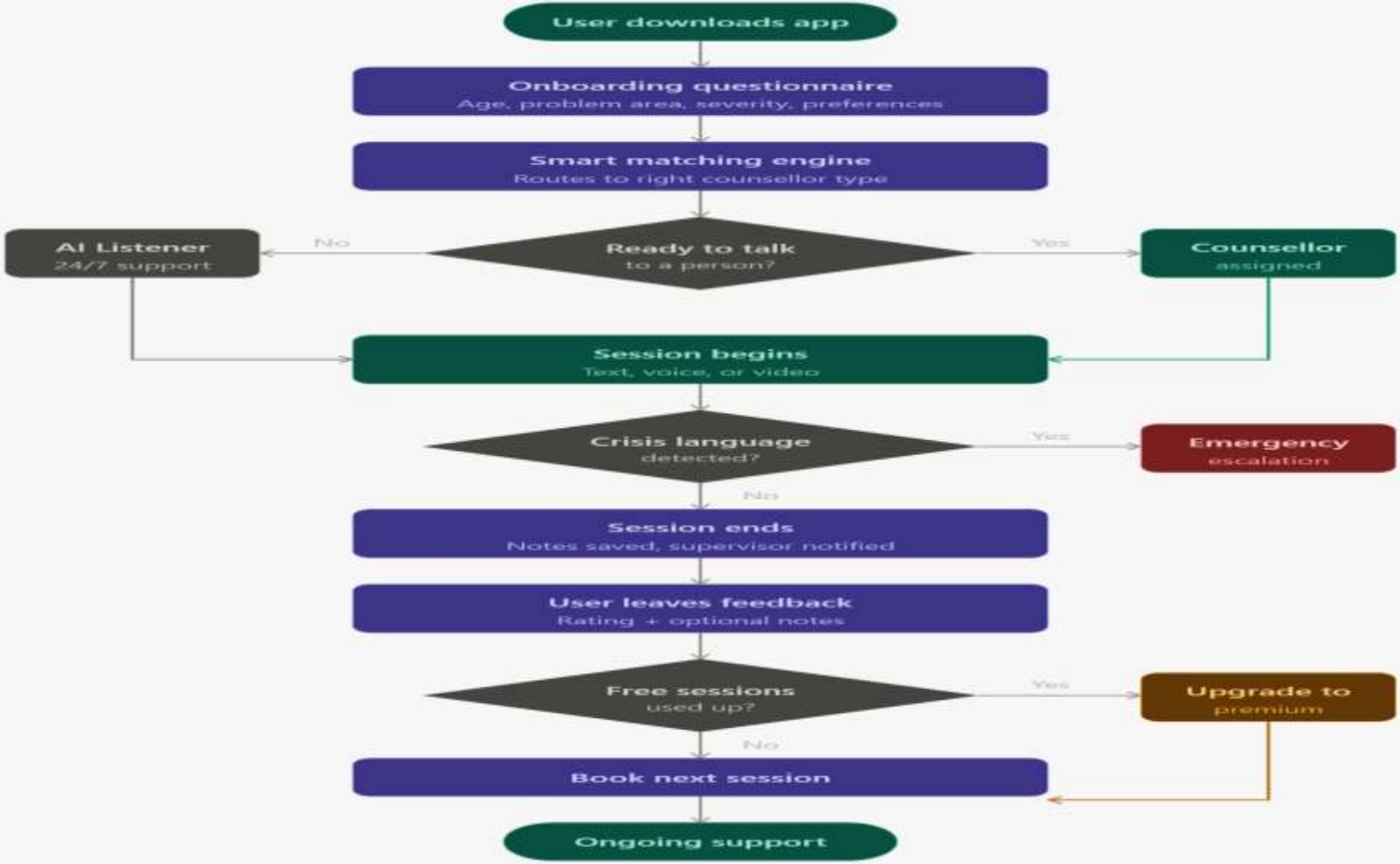
```
brydge-chatbot/
├── app.py              ← Flask server (run this)
├── listener.py         ← AI model + 220 frameworks system
prompt
├── requirements.txt    ← All Python packages
├── colab_setup.py      ← Copy-paste cells for Google
Colab
└── templates/
    └── chat.html       ← Full chat UI
```

`app.py`	Flask server — run this to start Brydge
`listener.py`	Mistral 7B + 220-framework system prompt
`chat.html`	Full chat UI with spheres, moods, quick replies
`colab_setup.py`	Copy-paste 6 cells → Brydge is live on Colab
`requirements.txt`	One pip install command for all libraries

How to Run Brydge — Free

1. **Open Google Colab**
   colab.research.google.com → Runtime → T4 GPU → Save
2. **Run Cell 1**
   pip install flask transformers bitsandbytes accelerate torch
3. **Run Cell 2**
   Downloads Mistral 7B MH (8–12 mins). No token needed.
4. **Run Cells 3–5**
   System prompt + chat function + Flask app all load
5. **Run Cell 6**
   localtunnel creates a public HTTPS URL. Brydge is live!
6. **Share the URL**
   Works on desktop, tablet, and mobile browser.

Total cost: £0 · No credit card · No GPU · All Apache 2.0 / MIT licensed


User downloads app
Onboarding questionnaire
Age, problem area, severity, preferences
Smart matching engine
Routes to right counsellor type
Ready to talk
to a person?
No
AI Listener
24/7 support
Yes
Counsellor
assigned
Session begins
Text, voice, or video
Crisis language
detected?
Yes
Emergency
escalation
No
Session ends
Notes saved, supervisor notified
User leaves feedback
Rating + optional notes
Free sessions
used up?
Yes
Upgrade to
premium
No
Book next session
Ongoing support

Why Brydge? — Comparison with ChatGPT & Other Apps

Feature	Brydge	ChatGPT	Woebot	Wysa
Mental health focused	✓	✗	✓	✓
220+ therapy frameworks	✓	✗	~	~
3 scoped life spheres	✓	✗	✗	✗
Crisis detection layer	✓	✗	~	✓
Judge LLM safety filter	✓	✗	✗	✗
Peer student counsellors	✓	✗	✗	✗
Free —	✓	✗	✗	✗
Privacy (no data sent to cloud)	✓	✗	~	~
Warm empathetic tone	✓	~	~	✓
Evidence-based responses	✓	~	~	~
Scope enforcement	✓	✗	~	✗

Brydge is the only free, scoped, evidence-based, crisis-safe AI psychologist that is also peer-supported and built for young people specifically.

Future Scope & Improvements

Human Counsellor Matching Engine

Connect users to trained masters year psychology students for real supervised sessions via text, voice, and video — the full Brydge vision.

Judge LLM Scoring Pipeline

A separate Llama 3.2 3B model that scores every AI response on 5 dimensions (empathy, safety, framework, helpfulness, tone) before it reaches the user.

Expand Life Spheres

Add grief, identity, career anxiety, relationships, financial stress, addiction awareness, chronic illness, and disability as new spheres beyond the current three.

Fine-tune on ESConv + CounselChat

QLoRA fine-tune Mistral 7B specifically on our 3 spheres using Google Colab's free T4 GPU. Model will dramatically improve on targeted conversations.

1,000 Situations Database

Build and integrate the full situation JSON database — 1,000 user problems mapped to frameworks, psychologists, and response strategies.

RLHF Improvement Loop

Reinforcement learning from human feedback

Psychology students rate AI responses weekly. Retrain on corrected examples. Brydge gets smarter every week automatically at zero extra cost.

Analytics Dashboard

Supervisor portal showing session quality scores, flagged conversations, common issues per sphere, and counsellor performance metrics.

Premium Tier

Free: 2 sessions/month with student counsellor. Premium: unlimited AI + priority matching + video sessions. Sustainable revenue model.

"Because every voice deserves to be heard."

"Healing starts by being heard."

- Free — built on open-source technology
- Evidence-based — 220+ frameworks from 8 core psychologists
- Safe — crisis detection before every response